6.3 Probability	understand the language of probability	Outcomes, equal likelihood, events, random
	understand and use the probability scale	P(certainty) = 1 P(impossibility) = 0
	understand and use estimates or measures of probability from theoretical models	
	understand the concepts of a sample space and an event, and how the probability of an event happening can be determined from the sample space	
	list all the outcomes for single events and for two successive events in a systematic way	For the tossing of two coins, the sample space can be listed as: Heads (H), Tails (T) (H, H), (H, T), (T, H), (T, T)
	estimate probabilities from previously collected data	
	calculate the probability of the complement of an event happening	P(not A) = 1 − P(A)

	use the addition rule of probability for mutually exclusive events	P(Either *A* or *B* occurring) = P(*A*) + P(*B*) when *A* and *B* are mutually exclusive
	understand and use the term *expected frequency*.	Determine an estimate of the number of times an event with a probability of $\frac{2}{5}$ will happen over 300 tries

Papers 3H and 4H (Higher Tier)

Content overview

Knowledge of the Foundation Tier content is assumed for students being prepared for the Higher Tier.

- Number
  - Numbers and the number system
- Algebra
  - Equations, formulae and identities
  - Sequences, functions and graphs
- Geometry
  - Shape, space and measure
  - Vectors and transformation geometry
- Statistics

Assessment overview

- Two written papers.
- Each paper is assessed through a two-hour examination set and marked by Edexcel.
- The total number of marks for each paper is 100.
- Each paper weighted at 50% of the qualification, targeted at grades A* – D.

Content

AO1 Number and algebra		
1 Numbers and the number system		
	Students should be taught to:	**Notes**
1.1 Integers	**See Foundation Tier.**	
1.2 Fractions	**See Foundation Tier.**	
1.3 Decimals	convert recurring decimals into fractions	$0.\dot{3} = \frac{1}{3}$, $0.2333\ldots = \frac{21}{90}$
1.4 Powers and roots	understand the meaning of surds	
	manipulate surds, including rationalising the denominator where the denominator is a pure surd	Express in the form $a\sqrt{2}$: $\frac{2}{\sqrt{8}}$, $\sqrt{18} + 3\sqrt{2}$ Express in the form $a + b\sqrt{2}$: $(3 + 5\sqrt{2})^2$
	use index laws to simplify and evaluate numerical expressions involving integer, fractional and negative powers	Evaluate: $\sqrt[3]{8^2}$, $625^{-\frac{1}{2}}$, $\left(\frac{1}{25}\right)^{\frac{3}{2}}$
	evaluate Highest Common Factors (HCF) and Lowest Common Multiples (LCM)	
1.5 Set language and notation	understand sets defined in algebraic terms	
	understand and use subsets	If A is a subset of B, then $A \subset B$
	understand and use the complement of a set	Use the notation A'
	use Venn diagrams to represent sets and the number of elements in sets	
	use the notation $n(A)$ for the number of elements in the set A	
	use sets in practical situations	
1.6 Percentages	use reverse percentages	In a sale, prices were reduced by 30%. The sale price of an item was £17.50. Calculate the original price of the item.
	repeated percentage change	Calculate the total percentage increase when an increase of 30% is followed by a decease of 20%
	solve compound interest problems	To include depreciation

1.7	**Ratio and proportion**	**See Foundation Tier.**	
1.8	**Degree of accuracy**	solve problems using upper and lower bounds where values are given to a degree of accuracy	The dimensions of a rectangle are 12 cm and 8 cm to the nearest cm. Calculate, to 3 significant figures, the smallest possible area as a percentage of the largest possible area.
1.9	**Standard form**	express numbers in the form $a \times 10^n$ where n is an integer and $1 \leq a < 10$ solve problems involving standard form	$150\ 000\ 000 = 1.5 \times 10^8$
1.10	**Applying number**	**See Foundation Tier.**	
1.11	**Electronic calculators**	**See Foundation Tier.**	

2 Equations, formulae and identities		
	Students should be taught to:	**Notes**